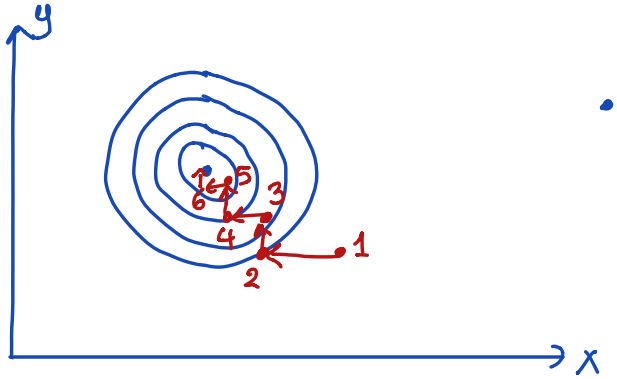
Multidimensional Optimization

(Ch. 14 in the book "Numerical Methods for Engineers")

Univariate Search Method

- Change one variable at a time to improve the approximation while the other variables are held constant.
- Problem reduces to a sequence of 1D searches

Graphically:

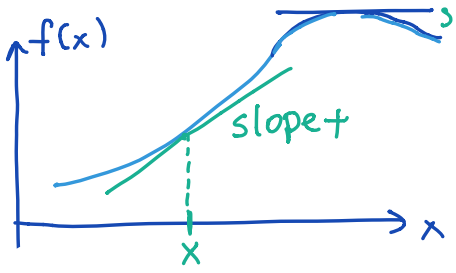


Start at pt. 1

- Move along x-axis with y constant
- Next move along y-axis with x constant
- Repeat the above 2 steps.

Gradient Methods

In 1D: First derivative of a 1D function provides a slope or tangent to the function.

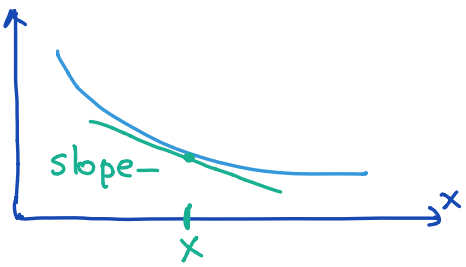


⇒ If slope > 0, increasing x will increase f(x)

⇒ If slope = 0, we are at optimum

$f'(x) = 0$

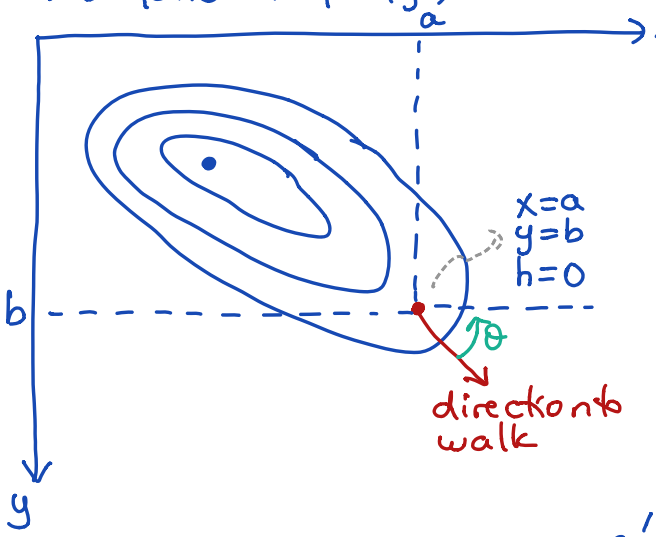
- if $f''(x) > 0 \Rightarrow$ a minimum (concave up)
- if $f''(x) < 0 \Rightarrow$ a maximum (concave down)



⇒ If slope < 0, decreasing x will increase f(x)

The Gradient

2D function $f(x,y)$



- You are on a mountain and want to reach the top
- You can move in arbitray direction
- What's the slope in that direction.
- The elevation along the direction (red arrow)

Directional derivative

$$g'(0) = \frac{\partial f}{\partial x}\cos\theta + \frac{\partial f}{\partial y}\sin\theta$$

evaluated at $x=a$, $y=b$

- The elevation along direction (red arrow) can be thought of a new function

$g(h)$. Slope at $x=a, y=b$

$g'(0)$

Goal: To gain the most elevation with the next step, what's the direction of steepest ascent?

Gradient: $\nabla f = \frac{\partial f}{\partial x}\underline{i} + \frac{\partial f}{\partial y}\underline{j}$ or $\nabla f = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}$

"del f"

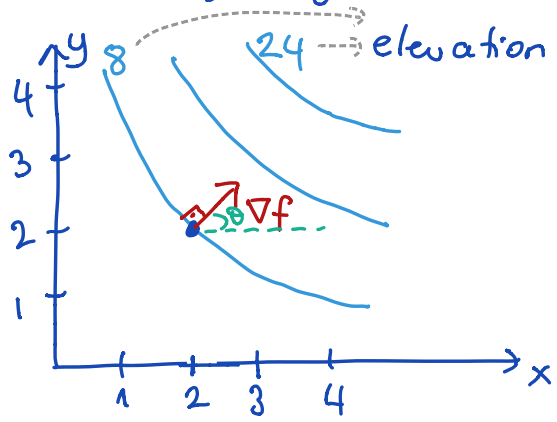
Generalize to n-dimensions:

$$\nabla f(\underline{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\underline{x}) \\ \frac{\partial f}{\partial x_2}(\underline{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\underline{x}) \end{bmatrix}$$

(*) Gradient tells us what direction to go locally for gaining max. elevation and how much we will gain by taking it.

EX Employ the gradient to evaluate the steepest ascent direction for the function:

$f(x,y) = xy^2$ at the point $(2,2)$



Gradient (∇f) is orthogonal to the elevation contour.

Sol Evaluate elevation:

$f(2,2) = 2(2)^2 = 8$

Find the partial derivatives:

$$\frac{\partial f}{\partial x} = y^2\Big|_{y=2} = 4 \qquad \frac{\partial f}{\partial y} = 2xy\Big|_{\substack{x=2\\y=2}} = 8$$

$$\text{Gradient} = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$

$\nabla f = 4\underline{i} + 8\underline{j}$, $\theta = \tan^{-1}\left(\frac{8}{4}\right) \cong 63^\circ$

slope in that direction: $\|\nabla f\| = \sqrt{4^2+8^2} = 8.944$

we will gain 8.944 units of elevation for a unit distance travelled along the steepest path (red arrow)

Note: As we move forward, the direction and magnitude of the steepest path will change. You can modify your climbing at each step.

The Hessian:

1D problems: Both first and second derivative provide valuable information for searching the optima.

$f'(x)=0 \Rightarrow$ we have reached an optimum

$f''(x)<0 \Rightarrow$ maximum is reached

$f''(x)>0 \Rightarrow$ minimum "

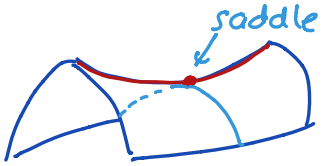
2D Functions:

(Hessian of f) $H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$

Determinant of Hessian: $|H| = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$

3 cases:

- If $|H|>0$ and $\frac{\partial^2 f}{\partial x^2}>0 \Rightarrow f(x,y)$ has a local minimum.
- If $|H|>0$ and $\frac{\partial^2 f}{\partial x^2}<0 \Rightarrow$ " " maximum.
- If $|H|<0$, then $f(x,y)$ has a saddle point. (neither max. or min)



Finite-Difference Approximations for Derivatives

$$\frac{\partial f}{\partial x} = \frac{f(x+\delta x, y) - f(x-\delta x, y)}{2\delta x}$$

δ: small perturbation

$$\frac{\partial f}{\partial y} = \frac{f(x, y+\delta y) - f(x, y-\delta y)}{2\delta y}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{f(x+\delta x, y) - 2f(x,y) + f(x-\delta x, y)}{\delta x^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{f(x, y+\delta y) - 2f(x,y) + f(x, y-\delta y)}{\delta y^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{f(x+\delta x, y+\delta y) - f(x+\delta x, y-\delta y) - f(x-\delta x, y+\delta y) + f(x-\delta x, y-\delta y)}{4\,\delta x\,\delta y}$$

- If possible use the analytical derivatives since they will be exact.

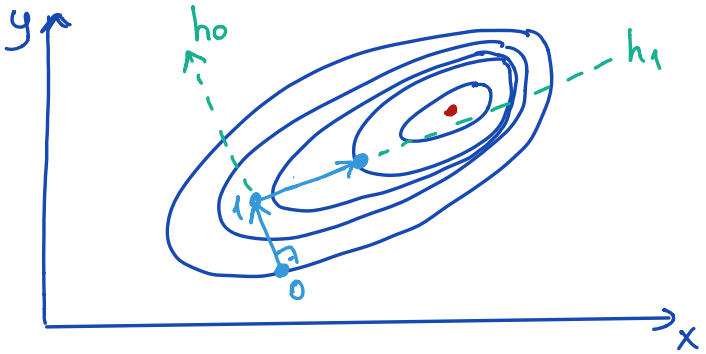
Steepest Ascent Method

- Determine the maximum slope at starting position.
- walk a short distance along the gradient direction.
- Stop, reevaluate the gradient and walk another short distance
- Repeat the proces and eventually get to the top of the hill.

→ Above approach is not very practical because of continuous re-evaluation of the gradient.

→ Preferred approach:

Move in a fixed path along the initial gradient until $f(x,y)$ stops increasing. Then repeat the process.



Start at pt. 0, move in the direction of gradient (ho) until a maximum is reached (pt. 1)

Problem:

(1) Determine the "best" direction to search.

(2) " " "best" value in that direction.

Transform $f(x,y)$ into a function of h along the gradient direction:

Start at (x_0, y_0) and move in the direction of h:

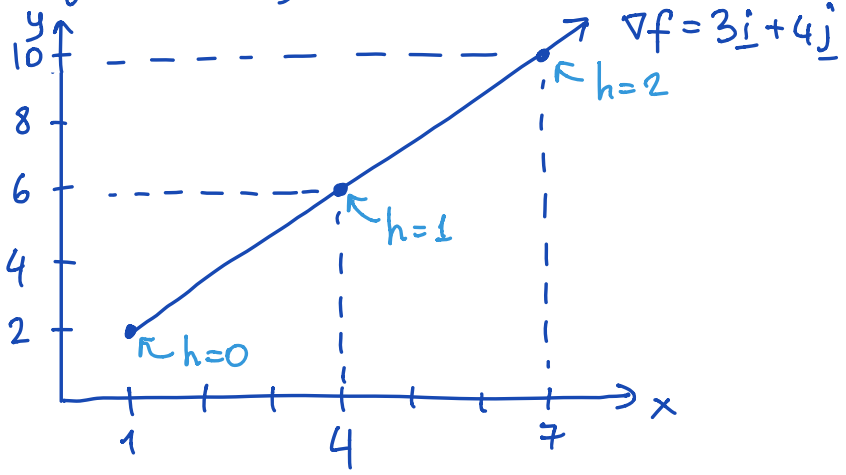
$$x = x_0 + \frac{\partial f}{\partial x} h$$

$$y = y_0 + \frac{\partial f}{\partial y} h$$

h = distance along h-axis.

Let $x_0 = 1$, $y_0 = 2$, $\nabla f = 3\underline{i} + 4\underline{j}$

$$\left.\begin{aligned} x &= 1 + 3h \\ y &= 2 + 4h \end{aligned}\right\}$$ Coordinates of any pt. along h-axis.



Ex $f(x,y) = 2xy + 2x - x^2 - 2y^2$

Develop a 1D version of the eqn along the gradient at point $x = -1$ and $y = 1$.

$$\frac{\partial f}{\partial x} = 2y + 2 - 2x \Big|_{\substack{x=-1 \\ y=1}} = 2 + 2 + 2 = 6$$

$$\frac{\partial f}{\partial y} = 2x - 4y \Big|_{\substack{x=-1 \\ y=1}} = -2 - 4 = -6$$

$x = -1 + 6h$

$y = 1 - 6h$

$f(x,y) = f(-1+6h, 1-6h)$

$= 2(-1+6h)(1-6h) + 2(-1+6h) - (-1+6h)^2 - 2(1-6h)^2$

$g(h) = -180h^2 + 72h - 7$

How far along this path do we travel?

until h^* max. of $g(h)$.

EX maximize $f(x,y) = 2xy + 2x - x^2 - 2y^2$

initial guess: $x = -1$, $y = 1$

Analytical solution:

$$\left.\begin{aligned} \frac{\partial f}{\partial x} &= 2y + 2 - 2x = 0 \\ \frac{\partial f}{\partial y} &= 2x - 4y = 0 \end{aligned}\right\} \text{Solve} \Rightarrow \left.\begin{aligned} x &= 2 \\ y &= 1 \end{aligned}\right\} \text{optimum pt.}$$

$$\frac{\partial^2 f}{\partial x^2} = -2 \qquad \frac{\partial^2 f}{\partial y^2} = -4 \qquad \frac{\partial^2 f}{\partial x \partial y} = 2$$

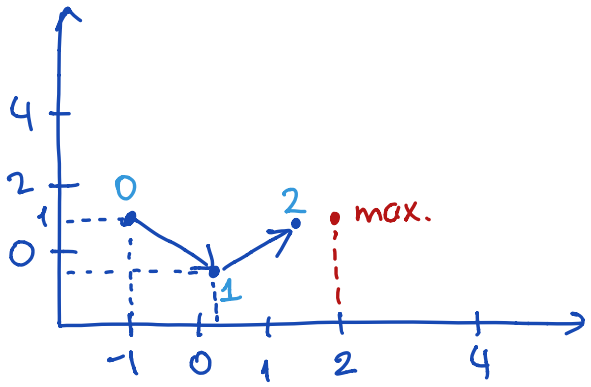
$|H| = -2(-4) - 2^2 = 4 > 0$ and $\frac{\partial^2 f}{\partial x^2} < 0 \Rightarrow f(2,1)$ is a maximum.

Using Steepest Ascent:

$g(h) = -180h^2 + 72h - 7$

$g'(h^*) = 0 \Rightarrow -360h + 72 = 0 \Rightarrow h^* = 0.2$

$$\left.\begin{aligned} x &= -1 + 6(0.2) = 0.2 \\ y &= 1 - 6(0.2) = -0.2 \end{aligned}\right\}$$



point 0 : (-1, 1)

point 1 : (0.2, -0.2)

point 2 : (1.4, 1)

opt.(max) : (2, 1)

Repeat the procedure:

$\left.\frac{\partial f}{\partial x}\right|_{pt1} = 1.2 \qquad \left.\frac{\partial f}{\partial y}\right|_{pt1} = 1.2$

$\nabla f = 1.2\underline{i} + 1.2\underline{j}$

$x = 0.2 + 1.2h$

$y = -0.2 + 1.2h$

$f(\underbrace{0.2+1.2h}_{x}, \underbrace{-0.2+1.2h}_{y}) = g(h) = -1.44h^2 + 2.88h + 0.2$

$g'(h^*) = 0 \Rightarrow -2.88h + 2.88 = 0$

$\Rightarrow h^* = 1$

$\Rightarrow x = 0.2 + 1.2(1) = 1.4$

$y = -0.2 + 1.2(1) = 1$ } pt.2